

Section – C

(Long-answer Type Questions)

3. Answer any **four** questions of the following :

$$10 \times 4 = 40$$

- (a) What is Deadlock ? Explain the four necessary conditions for a deadlock to occur.
- (b) Write a Shell Script to input a number and find its Factorial.
- (c) Explain the Banker's Algorithm for Deadlock Avoidance.
- (d) Write a Shell Script to generate the Fibonacci Series up to N terms.
- (e) Explain Page Replacement Algorithms with a simple reference string example.
- (f) Explain the following :
 - (i) Booting Process
 - (ii) Spooling
 - (iii) Dispatcher

2025

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the Sections as directed.*

(Operating System)

Section – A

(Objective Type Questions)

1. [A] Choose the correct answer of the following :

$$1 \times 5 = 5$$

(a) Which is not a type of Operating System ?

- (i) Batch (ii) Time-Sharing
- (iii) Real-Time (iv) Oracle

- (b) BIOS stands for :
- (i) Basic Input Output System
 - (ii) Binary Input Output System
 - (iii) Basic Internal Operating System
 - (iv) None of these
- (c) Which program acts as the core of the Operating System ?
- (i) Shell (ii) Kernel
 - (iii) Utility (iv) Compiler
- (d) Process Control Block (PCB) is a data structure used by :
- (i) User (ii) OS
 - (iii) Compiler (iv) CPU
- (e) Which command changes the owner of a file ?
- (i) Chmod (ii) Chown
 - (iii) Chgrp (iv) Cat

[B] Fill in the blanks : 1×5 = 5

- (a) A process created by the main process is called a _____.

- (b) The _____ is the core of the operating system that interacts directly with hardware.
- (c) _____ command is used to create a new directory in UNIX.
- (d) In vi editor, :q! command is used to quit without saving changes.
- (e) pwd command stands for _____.

Section – B

(Short-answer Type Questions)

2. Answer any **four** questions of the following in not more than **200** words each : 5×4 = 20
- (a) Differentiate between Kernel and Shell.
 - (b) Define Operating System and list its main functions.
 - (c) What is the difference between Multitasking and Multiprocessing ?
 - (d) Explain the significance of file permissions r, w and x in UNIX.
 - (e) Explain the SJF (Shortest Job First) scheduling algorithm.